

Solving Linear Equations

Recap

- We have established the importance and usefulness of matrix theory and linear algebra in data sciences
- Concepts covered previously
 - Data representation using matrices
 - Identifying linear relationships (if any) amongst attributes
- How do we establish these linear relationships?
 - Using null space
 - In order to calculate null space, we should be able to solve matrix equations
 - We will now focus on extracting solutions for matrix equations



SOLVING MATRIX EQUATIONS

Preliminaries

- We are considering the following data matrix A for identifying the solutions

$$A\mathbf{x} = \mathbf{b}$$

$$A(m \times n); \mathbf{x}(n \times 1); \mathbf{b}(m \times 1)$$

- Generalized linear equations can be represented in the above format.
- A generally represents the data matrix and $\mathbf{x}$ the attributes vector
- m and n are the number of equations and variables respectively.
- $\mathbf{b}$ is the general RHS used commonly

Categorization

$$m = n$$

- Data samples available equal to the number of attributes
- Easiest case to solve

$$m > n$$

- More data samples available than the number of attributes
- Usually, no unique solutions

$$m < n$$

- Lesser data samples available than the number of attributes
- Usually, multiple solutions

We look into these cases independently

Full row and column rank: Concepts

- Consider a matrix data matrix A ($m \times n$)

Full Row Rank

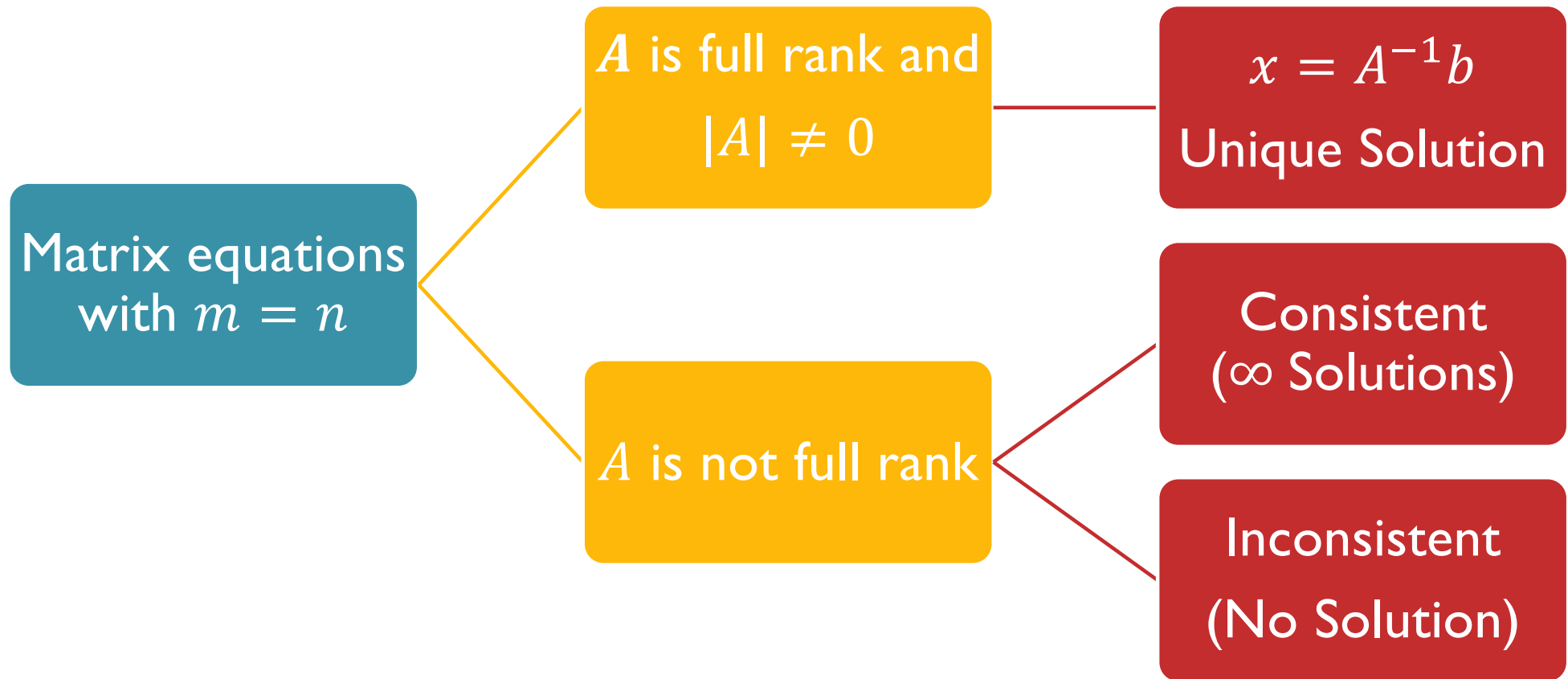
- When all the rows of the matrix are linearly independent
- Data sampling does not present a linear relationship – samples are independent

Full Column Rank

- When all the columns of the matrix are linearly independent
- Attributes are linearly independent

When A is such that $m = n$ then full row rank and full column rank are equivalent

Case 1: $m = n$



Case 1: Example 1.1

$$\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 7 \\ 10 \end{bmatrix}$$

$$|A| \neq 0$$

$$\text{rank}(A) = 2 = \text{no. of columns}$$

- This implies that A is full rank

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}^{-1} \begin{bmatrix} 7 \\ 10 \end{bmatrix} = \begin{bmatrix} -2 & 1.5 \\ 1 & -0.5 \end{bmatrix} \begin{bmatrix} 7 \\ 10 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

- Thus, the solution for the given example is $(x_1, x_2) = (1, 2)$

R Code

```
A=matrix(c(1,2,3,4),ncol=2, byrow=F)
b=c(7,10)
x=solve(A)%*%b
```

Console output

```
> x
     [,1]
[1,]    1
[2,]    2
```


Case 1: Example 1.2

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \end{bmatrix}$$

$$|A| = 0; \text{rank}(A) = 1; \text{nullity} = 1$$

- Checking consistency

$$\begin{bmatrix} x_1 + 2x_2 \\ 2x_1 + 4x_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \end{bmatrix}$$

$$\text{Row}(2) = 2\text{Row}(1)$$

R Code

```
A=matrix(c(1,2,2,4),ncol=2, byrow=F)
b=c(5,10)
x=solve(A)%*%b
```

- Thus, the equations are consistent with only 1 linearly independent equation
- The solution set for (x_1, x_2) is infinite because we have only 1 linearly independent equation and 2 variables

Output

```
x=solve(A)%*%b
```

```
Error in solve.default(A) : Lapack routine dgesv: system is exactly singular: U[2,2] = 0
```

Case 1: Example 1.3

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 9 \end{bmatrix}$$

$$\begin{aligned} |A| &= 0 \\ \text{rank}(A) &= 1 \\ \text{nullity} &= 1 \end{aligned}$$

R Code

```
A=matrix(c(1,2,2,4),ncol=2, byrow=F)
b=c(5,9)
x=solve(A)%*%b
```

- Checking consistency

$$\begin{bmatrix} x_1 + 2x_2 \\ 2x_1 + 4x_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 9 \end{bmatrix}$$

$$2\text{Row}(1) = 2x_1 + 4x_2 = 10 \neq 9$$

- Thus, the equations are inconsistent
- Hence, we cannot find a solution to (x_1, x_2)

Output

```
x=solve(A)%*%b
Error in solve.default(A) : Lapack routine dgesv: system
is exactly singular: U[2,2] = 0
```

Case 2: $m > n$

- This case addresses the problem of not enough attributes
- Since the number of equations is greater than the number of variables, not all equations can be satisfied
- It is hence, sometimes termed as a no-solution case
- But, we can view the same problem in an optimization perspective to gain some more insights

Case 2: An optimization perspective

- Instead of identifying a solution to $Ax - b = 0$ we can identify an x such that we $\min(Ax - b)$

- The same optimization can be reframed as

$$\begin{aligned} & \min[(Ax - b)^T (Ax - b)] \\ &= \min[(b^T - x^T A^T) (Ax - b)] \\ &= \min[(x^T A^T Ax - 2b^T Ax + b^T b) = f(x)] \end{aligned}$$

- Thus we observe that the optimization problem is a function of x . We can present the whole optimization problem as a differential

$$\nabla f(x) = 0$$

- The solution to this is thus obtained as follows

Case 2: An optimization perspective

- We can identify the solution to the optimization problem by differentiating f

$$2(A^T A)x - 2A^T b = 0$$

$$(A^T A)x = A^T b$$

$$\mathbf{x} = (A^T A)^{-1} A^T \mathbf{b}$$

- This is the solution we obtain to the second case using optimization concepts

Case 2: Example

$$\begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}$$

- We have the values of $\mathbf{A}$ and $\mathbf{b}$.
- $m = 3, n = 2$
- Using the optimization concepts,

$$\mathbf{x} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{b}$$
$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \left(\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}$$

Case 2: Example continued

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \left(\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \left(\begin{bmatrix} 14 & 3 \\ 3 & 1 \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0.2 & -0.6 \\ -0.6 & 2.8 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}$$

Case 2: Example continued

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0.2 & -0.6 \\ -0.6 & 2.8 \end{bmatrix} \begin{bmatrix} 20 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

- Thus, the solution for the given example is $(x_1, x_2) = (1, 2)$

R Code

```
A=matrix(c(1,2,3,0,0,1),ncol=2, byrow=F)
b=matrix(c(1,2,5),ncol=2, byrow=F)
x=inv(t(A)%*%A)%*%t(A)%*%b
x
```

Console output

```
> x=inv(t(A)%*%A)%*%t(A)%*%b
> x
      [,1]
[1,]    1
[2,]    2
>
```


Case 3: $m < n$

- This case addresses the problem of more than enough attributes
- Since the number of attributes is greater than the number of equations, we can obtain multiple solutions for the attributes
- It is hence, sometimes termed as an infinite-solution case
- What is the rationale for us to choose a single solution from the set of infinite possible solutions?

Case 3: An optimization perspective

- Pose an optimization problem for the parent equation $Ax = b$ such that

$$\min \left(\frac{1}{2} x^T x \right) \text{ s.t. } Ax = b$$

- Similar to the approach we took in case 2, we can define a function $f(x, \lambda)$

$$\min \left[f(x, \lambda) = \frac{1}{2} x^T x + \lambda^T (Ax - b) \right]$$

- Thus, we obtain the solution to be

$$x + A^T \lambda = 0$$

Case 3: An optimization perspective

- The simplification of the same leads to the following

$$x = -A^T \lambda$$

Pre-multiplying by A

$$b = -AA^T \lambda$$

Thus we obtain the value of λ as

$$\lambda = -(AA^T)^{-1}b$$

We obtain the solution for x

$$x = -A^T \lambda = A^T (AA^T)^{-1} b$$

- Note the similarities between the solutions obtained in cases 2 and 3

Case 3: Example

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

- We have the values of A and b.
- $m = 2, n = 3$
- Using the optimization concepts,

$$\begin{aligned} \mathbf{x} &= \mathbf{A}^T (\mathbf{A} \mathbf{A}^T)^{-1} \mathbf{b} \\ \mathbf{x} &= \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \left(\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \right)^{-1} \begin{bmatrix} 2 \\ 1 \end{bmatrix} \end{aligned}$$

Case 3: Example

$$x = \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 14 & 3 \\ 3 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$x = \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} -0.2 \\ 1.6 \end{bmatrix}$$

R Code

```
A=matrix(c(1,0,2,0,3,1),ncol=3)
b=c(2,1)
library(MASS)
x = t(A)%*%inv(A%*%t(A)) %*%b
x
```

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -0.2 \\ -0.4 \\ 1 \end{bmatrix}$$

Console output

```
A=matrix(c(1,0,2,0,3,1),ncol=3, byrow=F)
b=c(2,1)
x = t(A)%*%inv(A%*%t(A)) %*%b
x
```

- The solution for the given example is $(x_1, x_2, x_3) = (-0.2, -0.4, 1)$

Generalization

- The above described cases cover all the scenarios one might encounter while solving for the linear relationships amongst attributes using matrices.
- Is there any form in which the results obtained for cases 2 and 3 can be generalized?
- The concept we used to generalize the solutions is called as Moore-Penrose pseudo-inverse of a matrix
- The pseudo inverse is used as follows

$$Ax = b$$

The solution becomes

$$x = A^+ b$$

- Singular Value Decomposition can be used to calculate the pseudo inverse or the generalized inverse (A^+)

Two examples revisited

Example 2

R Code

```
A=matrix(c(1,2,3,0,0,1),ncol=2, byrow=F)
b=matrix(c(1,2,5),ncol=1, byrow=F)
library(MASS)
x= ginv(A)%*%b
```

Solution

```
> x [,1]
[1,] 1
[2,] 2
```

Example 3

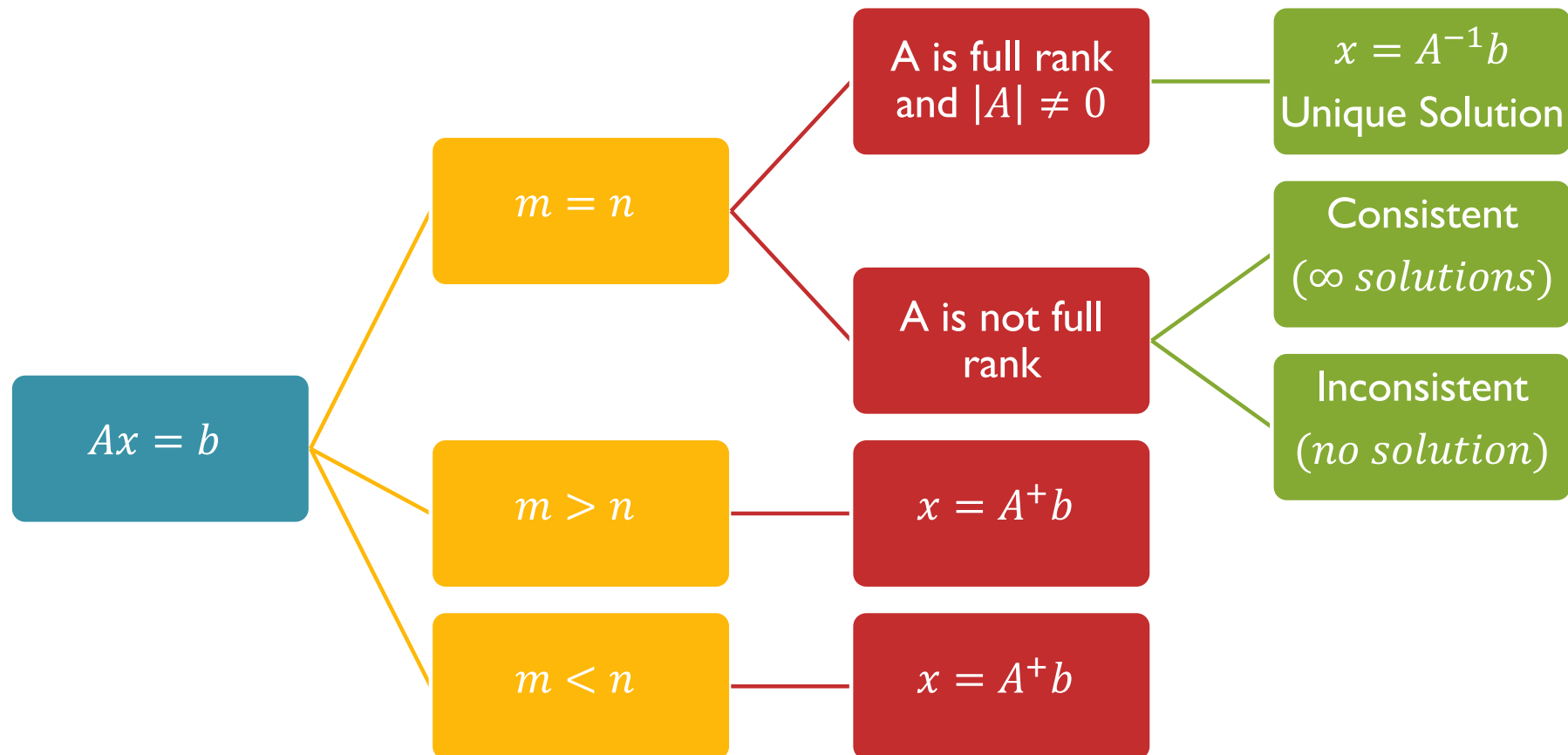
R Code

```
A=matrix(c(1,0,2,0,3,1),ncol=3, byrow=F)
b=c(2,1)
library(MASS)
x = ginv(A)%*%b
```

Solution

```
> x
      [,1]
[1,] -0.2
[2,] -0.4
[3,]  1.0
```

Summary



Where we define pseudo inverse A^+ appropriately